Project Report

❖ INTRODUCTION: -

1.1 Project Overview: -

Electricity consumption is a critical indicator of economic growth, industrial development, and living standards. This project analyzes electricity consumption patterns across 33 states and 5 regions of India for the period 2019–2020.
The dataset contains 16,599 records and includes state-wise monthly electricity usage data. The study particularly focuses on consumption variations before, during, and after the COVID-19 lockdown period, highlighting the economic and social impact on energy demand.
Using data preprocessing techniques and Tableau visualization, the project identifies regional trends, peak demand periods, and recovery patterns.

1.2 Purpose: -

The purpose of this project is to:

- Analyse electricity consumption trends over time.
- Compare regional demand patterns.
- Understand the impact of the COVID-19 lockdown on energy usage.
- Provide insights for data-driven energy planning and policy making.

❖ IDEATION PHASE: -

2.1 Problem Statement: -

The increasing demand for electricity due to urbanization and industrial growth has created challenges in energy management and sustainability. Electricity demand varies across regions and time due to industrial activity, seasonal effects, and socio-economic conditions. During the COVID-19 lockdown, significant disruptions occurred in industrial and commercial electricity consumption.
There is a need to analyse these variations systematically to understand consumption behaviour and post-lockdown recovery patterns.

2.2 Empathy Map Canvas: -

Thinks

- How can we optimize energy distribution?
- Are renewable sources sufficient?

Feels

- Concerned about environmental impact
- Overwhelmed by complex datasets

Speaks

- "We need better insights."
- "Data should guide policy decisions."

Does

- Reviews reports
- Analyses consumption statistics

Pain Points

- Scattered data sources
- No interactive comparison tools

Gains

- Clear trends
- Data-driven decisions

2.3 Brainstorming: -

Brainstorm, Idea Listing and Grouping - Build a regional comparison dashboard for electricity

- usage
- Identify seasonal consumption peaks
- Analyse sector-wise consumption (residential, commercial, industrial)
- Integrate population density data to correlate with energy use
- Create time-series graphs for trend analysis
- Add filtering options for dynamic insights
- Use color-coded maps to show consumption intensity
- Link weather and temperature data to electricity spikes
- Compare pre- and post-smart grid implementations
- Predict future consumption using historical data

❖ REQUIREMENT ANALYSIS: -

3.1 Journey Map: -

Stage	Phase Name	User Actions	Characteristics	Key Behaviour / Technology	Pain Points / Goals

1	Awareness & Initial Setup (Passive Phase)	Install traditional meters, pay monthly bills	Consumption is habitual, limited visibility, peak usage during routine hours	Traditional electricity meters	Surprise high bills, no control, no awareness of energy-saving opportunities
2	Data Visibility & Monitoring (Active Phase)	Install smart meters, use mobile apps, monitor real-time data	Real-time tracking of energy usage, awareness of peak hours	Smart meters (AMI), IoT devices, smart plugs, smart thermostats	Identifying high-consumption appliances, beginning energy-saving practices
3	Optimization & Smart Control (Prosumer Phase)	Automate appliances, use Smart Energy Management Systems	Load shifting to off-peak hours, automated scheduling	AI-driven optimization, smart scheduling, automated lighting & HVAC	Reducing costs, improving efficiency
4	Decentralized Energy Integration (Self-Sufficient Phase)	Install solar panels, battery storage, EV integration	Reduced grid dependence, self-consumption of renewable energy	Solar PV, Battery Energy Storage Systems (BESS), Electric Vehicles (EVs)	Selling excess energy, maximizing renewable use
5	Future-Ready Consumption (Sustainable Phase)	Participate in community energy sharing & DSM programs	Smart home integrated with decentralized green grid	Demand-Side Management (DSM), Smart Grid systems	Achieving net-zero, reducing carbon footprint

3.2 Solution Requirement: -

Functional Requirements:

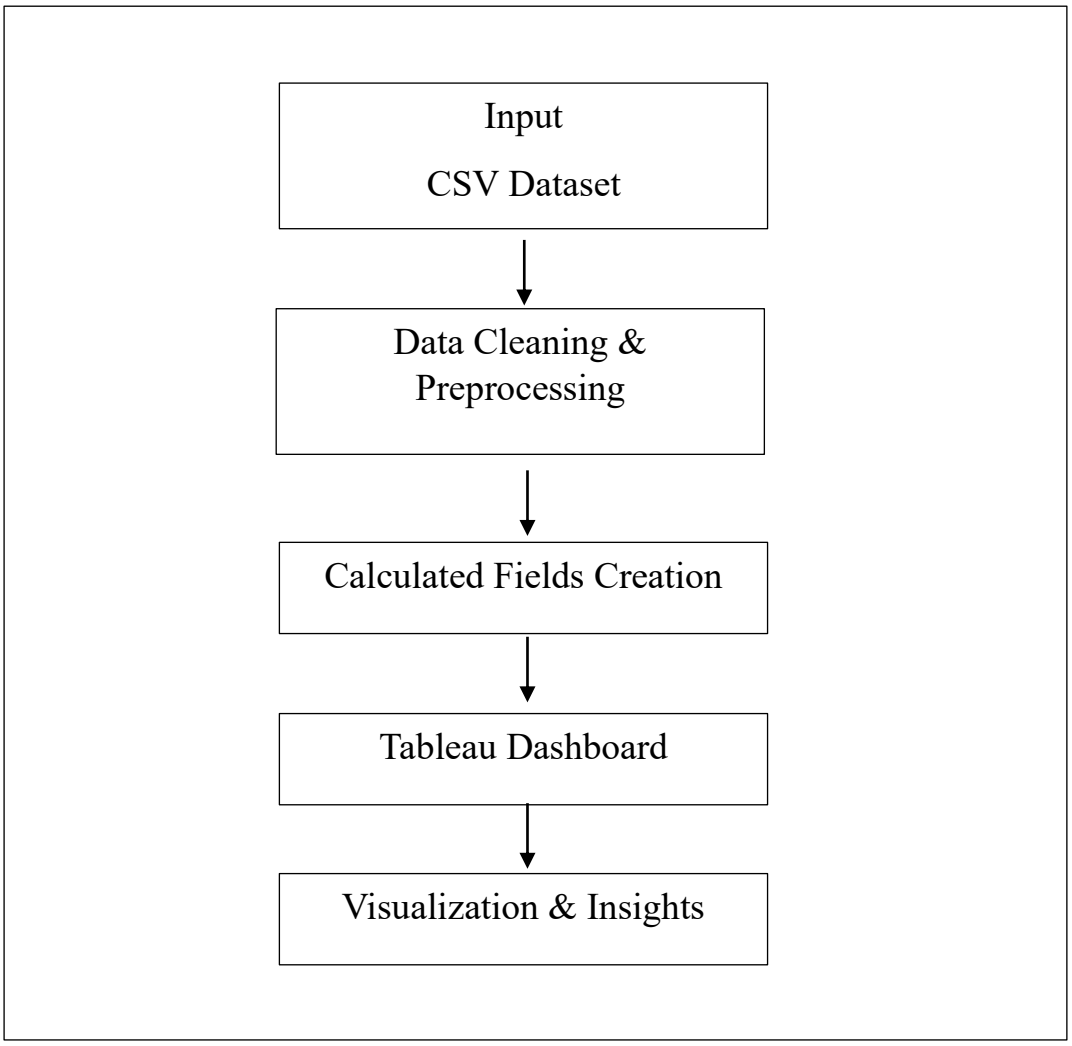
- Display month-wise electricity usage
- Compare regions and states
- Show year-over-year comparison
- Identify lockdown impact
- Provide interactive filters

Project Report

Non-Functional Requirements:

- Fast data rendering
- Accurate calculations
- User-friendly dashboard interface

3.3 Data Flow Diagram: -



3.4 Technology Stack: -

- Data Source: CSV Dataset
- Data Processing: Tableau
- Visualization: Tableau Dashboards & Story
- Documentation: PDF Report

Project Report

❖ PROJECT DESIGN: -

4.1 Problem Solution Fit: -

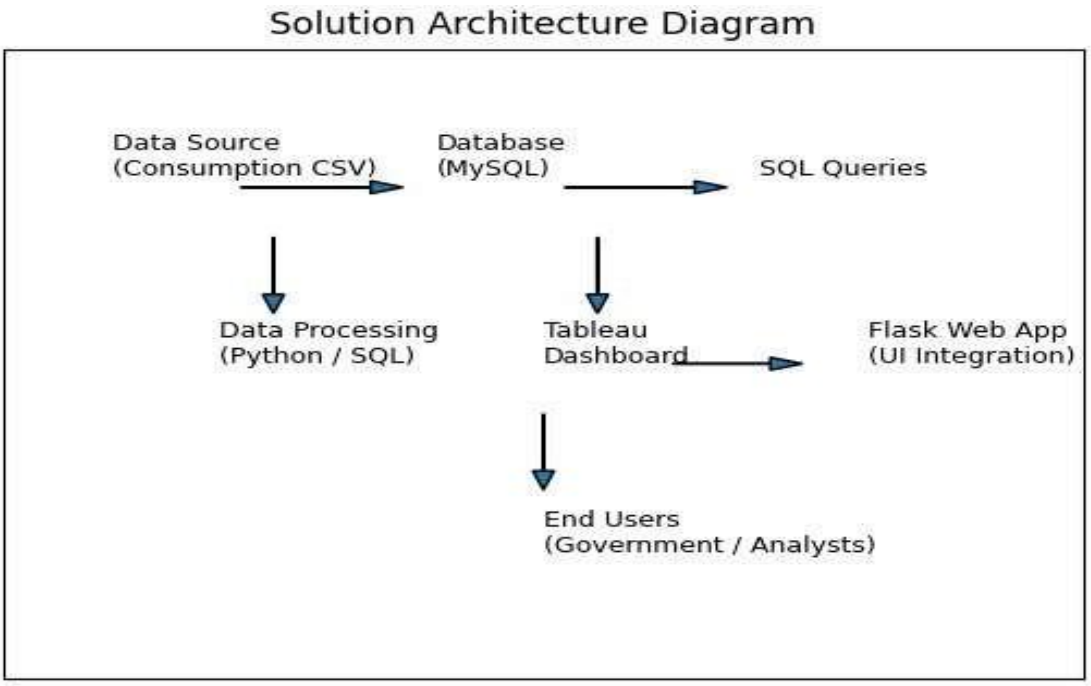
- The project effectively addresses the problem by transforming raw electricity consumption data into interactive dashboards that reveal trends and regional patterns.

4.2 Proposed Solution: -

A Tableau-based Electricity Consumption Analysis System that includes:

- Overall consumption trends
- Region-wise comparison
- Year comparison (2019 vs 2020)
- Lockdown impact analysis
- Top & Bottom states
- Quarter-wise usage

4.3 Solution Architecture: -



Project Report

❖ PROJECT PLANNING & SCHEDULING: -

5.1 Project Planning: -

Phase	Activity
Week 1	Dataset collection & review
Week 2	Data preprocessing
Week 3	Visualization creation
Week 4	Testing & documentation

❖ FUNCTIONAL AND PERFORMANCE TESTING: -

6.1 Performance Testing: -

Parameter	Status
Data Rendering	Successful
Filters Working	Yes
Year Comparison	Accurate
Region Mapping	Verified
Lockdown Analysis	Correctly Displayed
Dashboard Response Time	Fast

❖ RESULTS: -

Dashboard screenshots to be inserted here in final submission.

Project Report

❖ ADVANTAGES & DISADVANTAGES: -

Advantages: -

- Clear visualization of trends
- Helps in policy planning
- Identifies regional disparities
- Supports data-driven decisions

Disadvantages: -

- Limited to two years of data
- Does not include appliance-level consumption
- External factors beyond lockdown not deeply analyzed

❖ CONCLUSION: -

- The project demonstrates how electricity consumption patterns reflect economic activity and societal behaviour. The COVID-19 lockdown created an unprecedented decline in industrial energy demand, followed by gradual recovery.

- Through structured preprocessing and Tableau visualization, meaningful insights were generated from 16,599 records across 33 states.

- This analysis supports better energy forecasting and strategic planning for future disruptions.

❖ FUTURE SCOPE: -

- Include data for additional years
- Add predictive modelling using machine learning
- Integrate real-time electricity data
- Develop a web-based analytics dashboard
- Perform sector-wise (industrial, residential, commercial) breakdown

Project Report

❖ APPENDIX: -

Source Code:	app.py
	``python
	from flask import Flask, render_template
	
	app = Flask(__name__)
	
	@app.route("/")
	def home():
	    return render_template("home.html")
	
	@app.route("/dashboard")
	def dashboard():
	    return render_template("dashboard.html")
	
	@app.route("/story")
	def story():
	    return render_template("story.html")
	
	@app.route("/visualizations")
	def visualizations():
	    return render_template("visualizations.html")
	
	@app.route("/conclusion")
	def conclusion():
	    return render_template("conclusion.html")
	
	if __name__ == "__main__":
	    app.run(debug=True)
	``
	app.py

Dataset Link:	https://drive.google.com/drive/folders/1y5z43u5iTZnMzpMvNVv06wwTSwOlAUEI
GitHub Repository:	https://github.com/ROHIT4138/Plugging-into-the-Future-An-Exploration-of-Electricity-Consumption-Patterns-using-Tableau
Tableau Story	https://public.tableau.com/views/StoryonElectricityConsumption_17726386694010/StoryonElectricityConsumtion?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
Tableau Dashboard	https://public.tableau.com/views/Dashboard3_17722918268860/Dashboard3?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link